FDE Take-Home Assignment

Build a working AI agent with a real screen a sales team can use. Submit within 36 hours of getting this doc.

The problem

A small company sells packaging equipment to other businesses. It has 5 sales reps.

- Leads live in one Excel sheet and in WhatsApp chats with buyers.
- Reps forget to follow up, so good leads go cold.
- The owner can't see which leads are hot and who needs a reply today.

The owner wants one simple screen that tells each rep who to follow up with today, and what to say.

What to build

Build a lead follow-up agent with a working web app on top. Vibe coding is expected: use any AI coding tool you like.

Step 1. Make your own sample data

- At least 20 fake leads (name, company, phone, source, last contact date)
- At least 10 short WhatsApp-style chats between a rep and a buyer

Step 2. The agent must:

1. Read the leads and the chats
2. Tag each lead as Hot, Warm or Cold, with a one-line reason
3. Set a next follow-up date for each lead
4. Draft the next follow-up message for each lead

Step 3. The screen must:

- Show today's follow-ups at the top, most urgent first
- Make overdue follow-ups stand out clearly
- Let a rep open a lead and see the chat, the tag, the reason and the draft message
- Let a rep edit the draft and mark it as sent
- Let a rep add a new lead or a new chat, and have the agent update that lead

Out of scope: login, real WhatsApp sending, payments. Do not build these.

What to submit

Reply to the assignment email within 48 hours with these 4 things:

#	What	Details
1	Live link	A deployed app we can open and use. No local setup.
2	Code	A GitHub repo we can see
3	Video	Max 3 minutes. Show the app working as a sales rep would use it.
4	Short note	Half a page: what you built, how the agent decides Hot/Warm/Cold, what AI tools you used, and what you would build next

Missing any of the 4 means the submission is not reviewed.

Rules

- Any language, framework, AI model or AI coding tool is fine.
- Free tiers are fine for hosting and AI models.
- Work alone. You must be able to explain every part of what you built.
- Stuck on something? Pick the simplest option, write it down in your note, and keep going.

How we review it

We check each item as a pass or fail.

Check	Pass means
It works	The live link opens and the main flow runs with no errors
Agent is useful	Hot/Warm/Cold tags and reasons make sense for the chats
Drafts are usable	A rep could send the draft message with small edits or none
Screen is clear	A rep knows who to call today within 10 seconds of opening it
Scope is right	Only what was asked, done well. No half-built extras.
You can explain it	Your note and video clearly explain how it works and why

Pass 5 or more checks, and "It works" must be one of them, to move to the final round. In that round you walk us through your build life.